

TEST REPORT

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Beaverton, OR 97008
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2025 11 21 052 SU

Ordering Provider:
Rupa Health
Rajeshwari Nagaraja, MD
Reference Code
ord_j0p572j

Samples Received
11/21/2025
Report Date
12/01/2025

Samples Collected
Saliva - 11/19/25 05:45
Urine - 11/19/25 05:45
Urine - 11/14/25 13:30
Urine - 11/17/25 20:00
Urine - 11/14/25 01:56

Patient Name: Jennifer Pei Ru Hao
Patient Phone Number: 917 143 1266

Gender Female	Last Menses 10/17/2025	Height Unspecified	Waist Unspecified
DOB 2/15/1994 (31 yrs)	Menses Status Pre-Menopausal - Irregular	Weight Unspecified	

TEST NAME	RESULTS 11/14/25	RANGE
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Urinary Inhibitory Neurotransmitters

Tryptophan	 6021	2633-12688 µg/g Cr (Optimal 3970-8450)
Serotonin	 100.6	47.6-140.3 µg/g Cr (Optimal 61.0-103.2)
5-HIAA	 3605	2205-11816 µg/g Cr (Optimal 2988-5850)
GABA	 120 L	167-463 µg/g Cr (Optimal 193-367)
Glycine	 120	41-295 mg/g Cr (Optimal 61-159)
Taurine	 84.8	7.1-293.1 mg/g Cr (Optimal 24.5-134.1)

Urinary Excitatory Neurotransmitters

Glutamate	 3563	1213-4246 µg/g Cr (Optimal 1515-2710)
Glutamine	 60	27-106 mg/g Cr (Optimal 37-71)
Histidine	 84.4	10.8-98.9 mg/g Cr (Optimal 19.7-58.4)
Histamine	 215.6 H	3.6-44.3 µg/g Cr (Optimal 5.2-15.3)
N-Methylhistamine	 163	59-195 µg/g Cr (Optimal 79-140)
PEA	 26.9	3.6-38.8 µg/g Cr (Optimal 5.3-16.1)
Tyrosine	 7080	3128-15548 µg/g Cr (Optimal 4790-10278)
Tyramine	 622	187-910 µg/g Cr (Optimal 279-588)
Dopamine	 172	103-282 µg/g Cr (Optimal 144-240)
DOPAC	 3713 H	495-2456 µg/g Cr (Optimal 658-1449)
HVA	 17126 H	3025-9654 µg/g Cr (Optimal 3737-7048)

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The above results and comments are for informational purposes only and are not to be construed as medical advice. Please consult your healthcare practitioner for diagnosis and treatment.

David T. Zava

David T. Zava, Ph.D.
Laboratory Director

Alison McAllister, ND

Alison McAllister, ND.
(Ordering Provider unless
otherwise specified on page 1)

TEST NAME	RESULTS 11/14/25	RANGE
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Urinary Excitatory Neurotransmitters

Norepinephrine (pooled)	21.8	10.0-35.7 µg/g Cr (Optimal 15.0-28.1)
Normetanephrine	17.3	13.4-44.8 µg/g Cr (Optimal 17.9-31.7)
Epinephrine (pooled)	1.5	0.8-6.2 µg/g Cr (Optimal 1.4-4.2)
Ratio: Norepi/Epi	14.5	2.9-25.2 (Optimal 5.2-13.7)
VMA	3080	1996-5939 µg/g Cr (Optimal 2580-4766)

Urinary Inflammatory Markers

Kynurenine	412	108-1641 µg/g Cr (Optimal 257-960)
Kynurenic Acid	929	437-1719 µg/g Cr (Optimal 639-1200)
3-Hydroxykynurenine	188	80-822 µg/g Cr (Optimal 147-467)
Xanthurenic Acid	840	450-2175 µg/g Cr (Optimal 694-1510)

Urinary Creatinine

Creatinine (pooled)	0.94	0.3-2.0 mg/mL
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<dl = Less than the detectable limit of the lab. N/A = Not applicable; 1 or more values used in this calculation is less than the detectable limit. H = High. L = Low.

Therapies

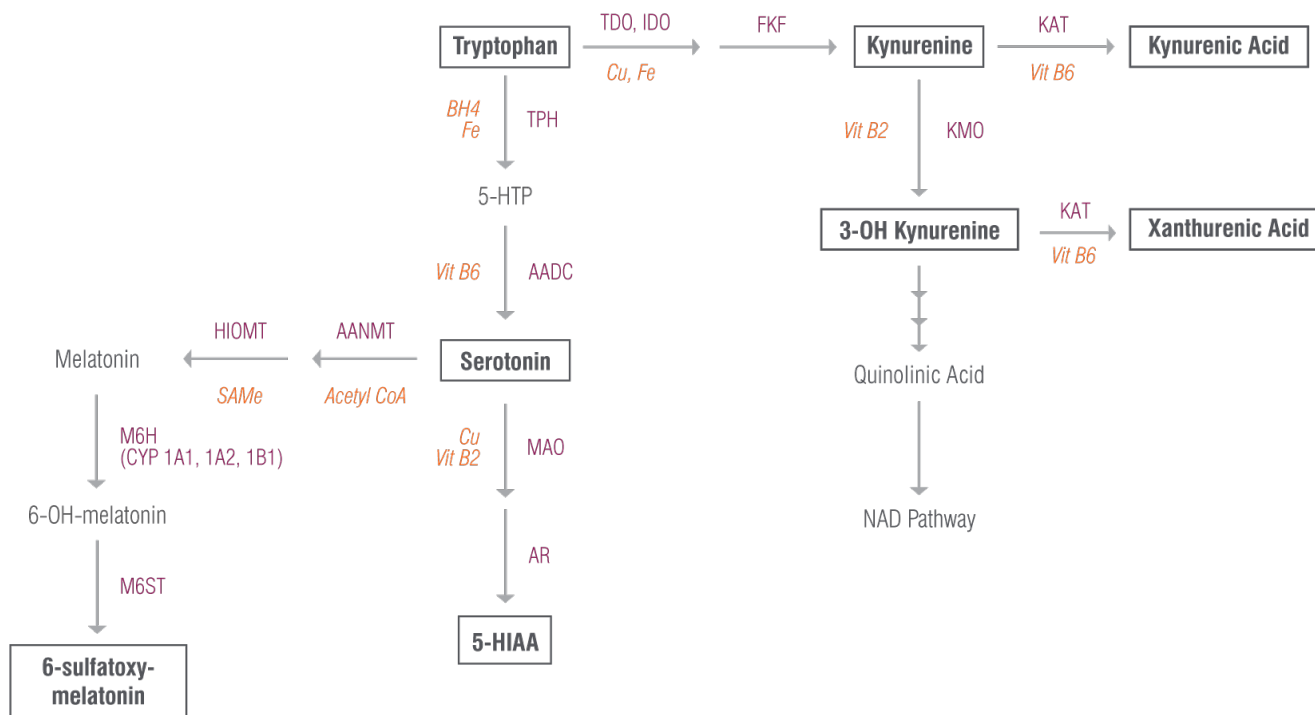
0.25mg BID oral Xanax (Pharmaceutical) (2 Days Last Used)

Disclaimer: Supplement type and dosage are for informational purposes only and are not recommendations for treatment. For a complete listing of reference ranges, go to www.zrtlab.com/reference-ranges.

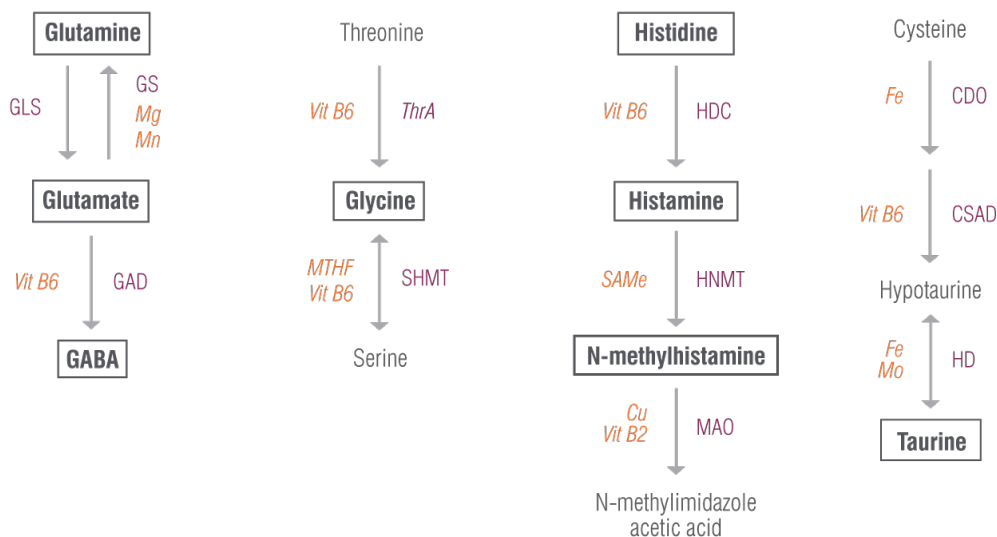
TEST NAME	WOMEN
Urinary Inhibitory Neurotransmitters	
Tryptophan	2633-12688 µg/g Cr (Optimal 3970-8450)
Serotonin	47.6-140.3 µg/g Cr (Optimal 61.0-103.2); 75-2137 µg/g Cr (w/5-HTP tx); 137.7-215.8 µg/g Cr 10-14y/o; 182.2-366.8 µg/g Cr <10y/o
5-HIAA	2205-11816 µg/g Cr (Optimal 2988-5850); 4008-114,204 µg/g Cr (w/5-HTP tx); 3802-9879 µg/g Cr 10-14y/o; 6331-18834 µg/g Cr <10y/o
GABA	167-463 µg/g Cr (Optimal 193-367); 197-311 µg/g Cr 10-14y/o; 270-633 µg/g Cr <10y/o
Glycine	41-295 mg/g Cr (Optimal 61-159); 97-267 mg/gr Cr 10-14y/o; 99-330 mg/gr Cr <10y/o
Taurine	7.1-293.1 mg/g Cr (Optimal 24.5-134.1)
Urinary Excitatory Neurotransmitters	
Glutamate	1213-4246 µg/g Cr (Optimal 1515-2710); 1669-2651 µg/g Cr 10-14y/o; 2157-4735 µg/g Cr <10y/o
Glutamine	27-106 mg/g Cr (Optimal 37-71)
Histidine	10.8-98.9 mg/g Cr (Optimal 19.7-58.4)
Histamine	3.6-44.3 µg/g Cr (Optimal 5.2-15.3); 12.4-20.5 µg/g Cr 10-14y/o; 14.5-38.4 µg/g <10y/o
N-Methylhistamine	59-195 µg/g Cr (Optimal 79-140)
PEA	3.6-38.8 µg/g Cr (Optimal 5.3-16.1); 3.7-10.8 µg/g Cr 10-14y/o; 6.8-32.3 µg/g Cr <10y/o
Tyrosine	3128-15548 µg/g Cr (Optimal 4790-10278)
Tyramine	187-910 µg/g Cr (Optimal 279-588)
Dopamine	103-282 µg/g Cr (Optimal 144-240); 257-395 µg/g Cr 10-14y/o; 382-770 µg/g Cr <10y/o
DOPAC	495-2456 µg/g Cr (Optimal 658-1449); 700-1428 µg/g Cr 10-14y/o; 1090-5203 µg/g Cr <10y/o
HVA	3025-9654 µg/g Cr (Optimal 3737-7048); 5382-9753 µg/g Cr 10-14y/o; 7295-21939 µg/g Cr <10y/o
Norepinephrine (pooled)	10.0-35.7 µg/g Cr (Optimal 15.0-28.1); 16.7-30.6 µg/g Cr 10-14y/o; 23.9-61.1 µg/g Cr <10y/o
Normetanephrine	13.4-44.8 µg/g Cr (Optimal 17.9-31.7); 27.0-50.1 µg/g Cr 10-14y/o; 24.9-70.1 µg/g Cr <10y/o
Epinephrine (pooled)	0.8-6.2 µg/g Cr (Optimal 1.4-4.2); 2.3-5.4 µg/g Cr 10-14y/o; 2.8-13.0 µg/g Cr <10y/o
Ratio: Norepi/Epi	2.9-25.2 (Optimal 5.2-13.7); 2.4-10.9 <15y/o
VMA	1996-5939 µg/g Cr (Optimal 2580-4766); 3570-5373 µg/g Cr 10-14y/o; 4310-9207 µg/g <10y/o
Urinary Inflammatory Markers	
Kynurenine	108-1641 µg/g Cr (Optimal 257-960)
Kynurenic Acid	437-1719 µg/g Cr (Optimal 639-1200)
3-Hydroxykynurenine	80-822 µg/g Cr (Optimal 147-467)
Xanthurenic Acid	450-2175 µg/g Cr (Optimal 694-1510)
Urinary Creatinine	
Creatinine (pooled)	0.3-2.0 mg/mL

Neurotransmitter Cascades

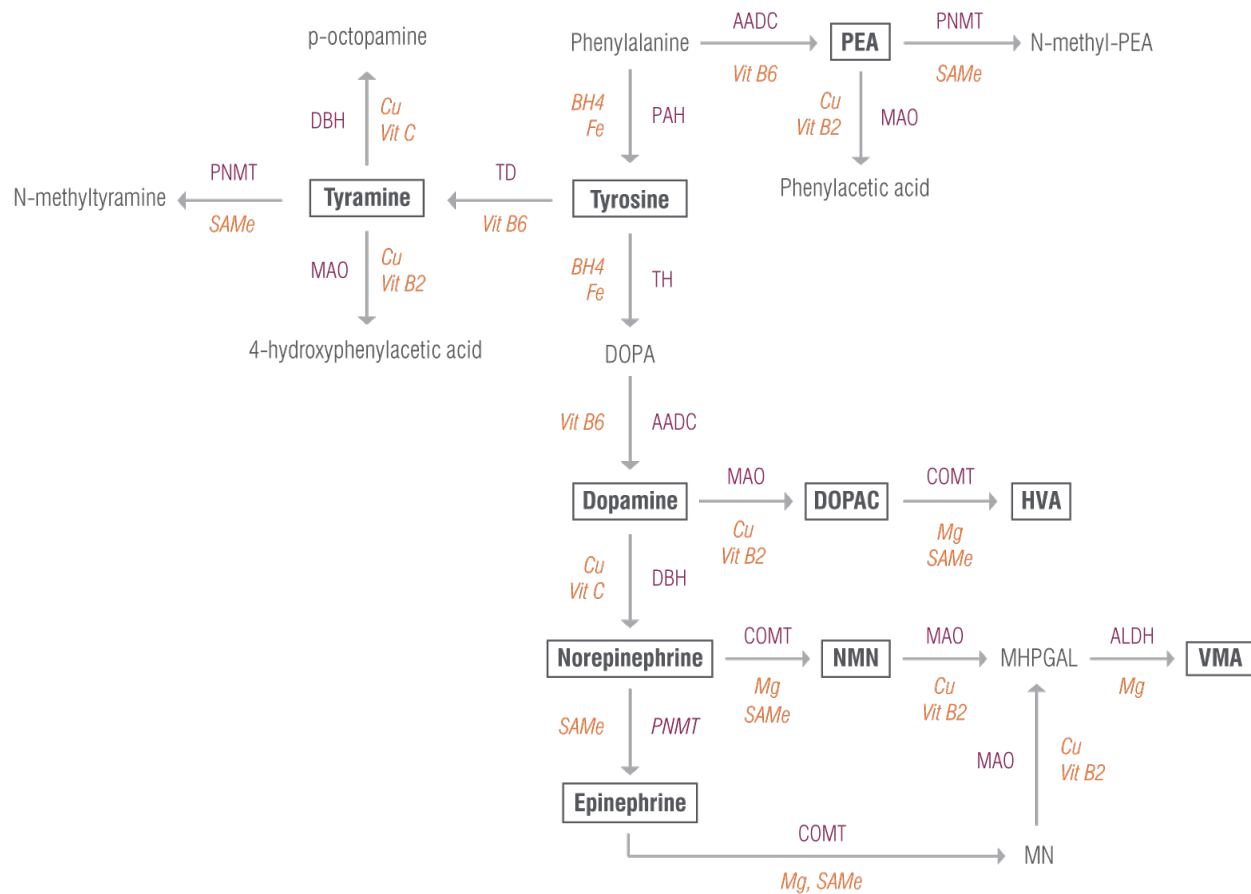
Tryptophan & Metabolites



Glutamate/GABA, Glycine, Histamine & Taurine



Catecholamines & Metabolites



Abbreviations & Key

Neurotransmitters
& Metabolites:

HVA	homovanillic acid
NMN	normetanephrine
PEA	phenethylamine
VMA	vanillylmandelic acid
5-HIAA	5-hydroxyindole 3-acetic acid

Cofactors:

BH4	tetrahydrobiopterin
Cu	copper
Fe	iron
Mg	magnesium
Mn	manganese
Mo	molybdenum
MTHF	methyltetrahydrofolate
S-AdoMet	S-adenosyl methionine

Enzymes:

AADC	aromatic L-amino acid decarboxylase
AANMT	arylalkylamine N-methyltransferase
ALDH	aldehyde dehydrogenase
AR	aldehyde reductase
CDO	cysteine dioxygenase
COMT	catechol-O-methyltransferase

CSAD	cysteinesulfonic acid decarboxylase
DBH	dopamine beta hydroxylase
FKF	N-Formyl kynurenine formamidase
GAD	glutamate decarboxylase
GLS	glutaminase
GS	glutamine synthetase
HD	hypotaurine dehydrogenase
HDC	histidine decarboxylase
HIOMT	hydroxyindole-O-methyltransferase
HNMT	histamine N-methyltransferase
IDO	indoleamine 2,3-dioxygenase
KAT	kynurenine aminotransferase
KMO	kynurenine hydroxylase/monooxygenase
MAO	monoamine oxidase
M6H	melatonin 6 hydroxylase
M6ST	melatonin 6 sulfotransferase
PAH	phenylalanine hydroxylase
PNMT	phenylethanolamine N-methyltransferase
SHMT	serine hydroxymethyltransferase
TD	tyrosine decarboxylase
TDO	tryptophan 2,3-dioxygenase
TH	tyrosine hydroxylase
ThrA	threonine aldolase
TPH	tryptophan hydroxylase

Disclaimer: Symptom Categories below show percent of symptoms self-reported by the patient compared to total available symptoms for each category. For detailed information on category breakdowns, go to www.zrtlab.com/patient-symptoms.

SYMPTOM CATEGORIES		RESULTS 11/14/25	
Estrogen / Progesterone Deficiency	28%		
Estrogen Dominance / Progesterone Deficiency	15%		
Low Androgens (DHEA/Testosterone)	24%		
High Androgens (DHEA/Testosterone)	30%		
Low Cortisol	32%		
High Cortisol	46%		
Hypometabolism	25%		
Metabolic Syndrome	38%		

SYMPTOM CHECKLIST	MILD	MODERATE	SEVERE
Aches and Pains			
Acne			
ADD/ADHD			
Addictive Behaviors			
Allergies			
Anxious			
Autism Spectrum Disorder			
Bleeding Changes			
Blood Pressure High			
Blood Pressure Low			
Blood Sugar Low			
Body Temperature Cold			
Bone Loss			
Breast Cancer			
Breasts - Fibrocystic			
Breasts - Tender			
Chemical Sensitivity			
Cholesterol High			
Constipation			
Depressed			
Developmental Delays			
Eating Disorders			
Fatigue - Evening			
Fatigue - Morning			
Fibromyalgia			
Foggy Thinking			
Goiter			
Hair - Dry or Brittle			
Hair - Increased Facial or Body			
Hair - Scalp Loss			
Headaches			
Hearing Loss			
Heart Palpitations			
Hoarseness			
Hot Flashes			
Incontinence			
Infertility			
Irritable			
Libido Decreased			
Mania			

SYMPTOM CHECKLIST		MILD	MODERATE	SEVERE
Memory Lapse				
Mood Swings				
Muscle Size Decreased				
Nails Breaking or Brittle				
Nervous				
Night Sweats				
Numbness - Feet or Hands				
OCD				
Panic Attacks				
PreMenstrual Dysphoric Disorder				
Pulse Rate Slow				
Rapid Aging				
Rapid Heartbeat				
Skin Thinning				
Sleep Disturbed				
Stamina Decreased				
Stress				
Sugar Cravings				
Sweating Decreased				
Swelling or Puffy Eyes/Face				
Tearful				
Triglycerides Elevated				
Urinary Urge Increased				
Uterine Fibroids				
Vaginal Dryness				
Water Retention				
Weight Gain - Hips				
Weight Gain - Waist				

Lab Comments

INHIBITORY NEUROTRANSMITTERS

TRYPTOPHAN

Tryptophan is within reference range. The essential amino acid tryptophan originates in diet and serves as a constituent of proteins and a precursor to neurotransmitters. Only a fraction of tryptophan is used by the GI tract, the vast majority of this amino acid enters portal circulation and undergoes liver metabolism. The remaining tryptophan pool, together with its liver degradation products, is distributed to peripheral circulation and transported to tissues such as the brain, heart, and skeletal muscle. Tryptophan not taken up by the upper GI tract is metabolized by resident microbiota.

Tryptophan is a substrate for two important biosynthetic pathways relevant to the neuropsychiatric interface: the generation of the neurotransmitter serotonin and therefore hormone melatonin, and the formation of kynurenine derivatives and therefore niacin (vitamin B3).

SEROTONIN

Serotonin is within reference range. Serotonin has calming effects and contributes to the feelings of well-being. Serotonin elevates mood, decreases anxiety, appetite, and libido, improves sleep and memory, eases depression, and helps regulate body temperature. Most of serotonin in the human body is produced in the gastrointestinal tract, where it stimulates gut motility.

5-HIAA

5-hydroxyindoleacetic acid (5-HIAA) is within reference range. 5-HIAA is the primary metabolite of serotonin via the actions of monoamine oxidase and aldehyde dehydrogenase enzymes.

GABA

GABA is below the optimal range. The brain's major inhibitory neurotransmitter, GABA functions as the "off" switch in the brain. GABA is essential to limiting brain neuron excitation so that input signals are balanced and not overdone. Appropriate levels of GABA prevent anxiety, improve mood, promote sleep, lower blood pressure, act as a muscle relaxant, aid in formation and storage of fear memories, increase insulin secretion and decrease blood glucose levels.

Research on urinary levels of GABA is scarce, however in individuals with anxiety and depression, GABA levels are low in the blood, in cerebrospinal fluid and in the brain (Mann, et. al. 2014; Goddard, 2016). The inhibitory and excitatory balance between GABA and glutamate is

very important for healthy brain function, and imbalance in these systems may contribute, in part, to the pathology of anxiety and depression, but we have yet to understand the mechanism.

THERAPEUTIC CONSIDERATIONS: with low GABA, supplementation with GABA, L-theanine, cofactor support (e.g. B6), growth hormone-releasing hormone, Ginkgo biloba, Ashwagandha, Kava, Valerian root, Melissa off. (lemon balm), Scutellaria sinensis (skullcap), Gotu Cola, Magnolia and Phellodendron bark, and probiotics may be helpful (Alramadhan et al., 2012; Awad et al., 2007; Alexeev et al., 2012; Dhakal et al., 2012). Additionally, yoga (Streeter et al., 2012) and meditation (Guglietti et al., 2013) increase brain GABA levels.

GLYCINE

Glycine is within normal range. Glycine is a simple, nonessential (can be made in the body) amino acid that plays a role in the production of DNA, phospholipids, collagen, creatine, heme and glutathione. Glycine serves as a neurotransmitter that modulates excitatory signals in the brain, and as an anti-inflammatory agent that calms aggression, improves sleep quality, stabilizes blood sugar, and improves metabolic parameters.

TAURINE

Taurine is within range. Taurine is a semi-essential or conditionally essential sulfur-containing amino acid and an inhibitory (calming) neurotransmitter. Taurine improves sleep, relieves anxiety, alleviates fatigue, aids with metabolism and digestion, and promotes glucose control and electrolyte balance.

The main source of taurine is diet. Taurine protects healthy cells and tissues, functions as a potent antioxidant to reduce oxidative stress, mitigates mitochondrial and endoplasmic reticulum stress, inhibits lipid peroxidation, improves energy metabolism, regulates gene expression, and participates in detoxification, calcium homeostasis and osmoregulation processes. By fulfilling all these functions, taurine is therefore protective in cardiovascular health, improves lean body mass and exercise performance. With regard to brain health, taurine serves a neuroprotective role, promotes neural development in embryonic and adult brain tissues, and is an important factor in neurogenesis.

EXCITATORY NEUROTRANSMITTERS

GLUTAMATE

Glutamate is above the optimal range. Research shows that in depressed individuals (self-reported), glutamate is elevated in urine (Jones, et. al. 2005), in blood (Ogawa, et. al. 2018), and in the brain (Sanacora, et. al. 2005). Elevated glutamate levels in depression are thought to be related to higher levels of inflammation (Muller, 2007) and inadequate mitochondrial function to properly metabolize glutamate (Abdallah, et. al. 2014). The brain's major excitatory neurotransmitter glutamate (also known as glutamic acid) functions as the "on" switch in the brain. Glutamate regulates appetite, thinking, increases gut motility, optimizes learning, modulates memory, mood and perception of pain, improves libido, and decreases sleep. Symptoms of high glutamate may include anxiety, depression, impulsivity, decreased focus, and insomnia. Additional studies on urinary glutamate levels report higher levels in patients with celiac disease (Marko, et. al. 1960), hyperthyroidism (Belanger, et. al. 1972), and rheumatoid arthritis (Jones, et. al. 2005).

THERAPEUTIC CONSIDERATIONS: Therapies such as GABA, L-theanine, lithium orotate and taurine may be beneficial to counter glutamate actions. NAC and Vitamin E can help reduce oxidative stress and co-factor supplementation (B3 and B6, magnesium, NAC) can aid in glutamate metabolism.

GLUTAMINE

Glutamine is within range. Glutamine is an essential and the most abundant free amino acid in the human body. Glutamine provides fuel for rapidly dividing cells (lymphocytes, enterocytes and epithelial cells of the intestines), helps balance ammonia levels in the body, improves immune system function, contributes to biosynthesis of proteins, amino acids, nucleic acids and glutathione, and protects intestinal lining. Additionally, glutamine increases glutamate and GABA levels in the brain and in the body.

HISTIDINE

Histidine is high-normal (>80th percentile), likely due to supplementation or recent dietary intake. Histidine is a semi-essential amino acid that gives rise to the neurotransmitter histamine. Histidine protects neurons, assists with making new blood cells, reduces inflammation and oxidative stress, helps with tissue repair and growth. Histidine also helps ameliorate fatigue, promotes clear thinking and concentration, reduces appetite, decreases anxiety, improves sleep and glucose homeostasis.

Research shows that histidine excretion is high with histidine administration (Du, Sun et al. 2017), in histidinemia (Auerbach, DiGeorge et al. 1962), and in diabetic nephropathy (Shao, Lu et al. 2020).

THERAPEUTIC CONSIDERATIONS: because high histamine has not been linked to adverse symptoms, no therapy is thought to be needed.

HISTAMINE

Histamine is above the optimal range, which may be due to allergies (self-reported). Research shows that individuals with gastrointestinal food allergies have higher urinary histamine levels (Raithel, et. al. 2015). This study showed that histamine was elevated in both IgE- and non-IgE food allergy types during consumption of offending foods, but not during the hypoallergenic diet.

This individual concurrently reports having ADHD. Literature suggests that some individuals with ADHD may have food sensitivities, and when present in diet, these foods could exacerbate symptoms of the disorder. Restriction/elimination diets, excluding artificial food coloring and excess sugar from diet have been shown by multiple studies to improve symptoms of hyperactivity and inattentiveness (Nigg and Holton, 2014).

Histamine plays a dual role in the body as a neurotransmitter and a modulator of the immune system that has anti-pain properties, plays a neuroprotective role in the brain, and contributes to optimal maintenance of cognition and memory. Histamine stimulates wakefulness and decreases sleep, stimulates gastric acid production, increases metabolism, suppresses appetite, and prevents weight gain. Histamine is a potent vasodilator and a pro-inflammatory agent. Additional research studies show that urinary histamine is elevated in patients with flushing disorder (Myers et al., 1981), cystitis (el-Mansoury et al., 1994), polycythemia (Horakova et al., 1977), and pregnancy (Harrison et al., 1974). Clinically, high histamine levels are implicated in depression, headaches, sensitivity to chemicals, and sleep difficulties.

THERAPEUTIC CONSIDERATIONS: Beneficial therapeutic strategies to reduce histamine levels may involve antihistamines and/or a low histamine diet. High histamine foods include but are not limited to beer, champagne, aged cheeses, eggplant, canned fish, fermented meat, red and white wine, sauerkraut, and spinach (Maintz and Novak, 2007), however individual food sensitivities ought to be considered as well. Additionally, flavonoids (green tea extract, quercetin, grape seed extract, ginkgo biloba, citrus bioflavonoids, bilberry extract, hawthorn extract) may be beneficial to ease the symptoms of high histamine (Murray et al., 2005).

N-METHYLHISTAMINE

N-Methylhistamine is high-normal (>80th percentile). N-Methylhistamine is a major metabolite of the neurotransmitter histamine. Research shows that N-methylhistamine excretion is elevated with high protein diet (Keyzer, Breukelman et al. 10/1984), in gastrointestinal food allergies (Raithel, Hagel et al. 2015), in irritable bowel disease (Winterkamp, Weidenhiller et al. 2002), in colitis (Schwab, Hahn et al. 2003), in histidinemia (Imamura, Watanabe et al. 1984), in chronic urticaria (Patel and Divekar 2017), in angioedema (Patel and Divekar 2017), and with interstitial cystitis (el-Mansoury, Boucher et al. 8/1994).

THERAPEUTIC CONSIDERATIONS: evaluation for food allergy sources may be warranted.

PEA

PEA is higher than the optimal range, which, most commonly is due to the use of amphetamine medication, that works to increase PEA (Kusaga et al., 2002; Zemetkin et al., 1984), however this type of medication was not indicated on the requisition form. PEA, also known as phenethylamine, easily crosses the blood brain barrier and acts as a "neuro-amplifier" - increasing the actions of dopamine (for wellbeing and feeling pleasure), norepinephrine (the brain's stimulant for wakefulness, alertness and higher performance), acetylcholine (for improving memory and mental activity), and serotonin (for better mood emotion and impulse control) (Paterson, et. al. 1990).

PEA is a biomarker for ADHD - urinary PEA levels are typically lower in patients with ADHD (self-reported) and increase with positive response to treatment. However, it's not unusual to see elevated PEA in patients that concomitantly present with mood issues, such as anxiety and insomnia (self-reported) (DeLisi, et. al., 1984). Therapeutic strategies to lower PEA may be beneficial to alleviate the self-reported symptoms of anxiety and insomnia.

Elevated urinary PEA levels have also been reported in patients on tricyclic antidepressants (Thornton, 1980), with bipolar disorder (Karoum et al., 1982) and with phenylketonuria (Reynolds et al., 1978).

THERAPEUTIC CONSIDERATIONS: consider evaluation of current medication for sources of pharmaceuticals or supplements that can increase PEA (e.g. phenylalanine). Therapeutic strategies that employ vitamin B2 supplementation, SAME or nutritional lithium may aid to lower PEA levels by promoting metabolism.

TYROSINE

Tyrosine is within range. Tyrosine is obtained from diet (sesame seeds, cheese, soy, meat, nuts and fish) or synthesized in the body from the amino acid phenylalanine. Tyrosine serves as a constituent of proteins and gives rise to neurotransmitters, like dopamine, norepinephrine and epinephrine; and the trace-amine tyramine. Additionally, in the thyroid gland, tyrosine can also be iodinated to give rise to thyroid hormones. Tyrosine enhances cognitive performance, energy, and alertness, and improves memory after sleep deprivation. Tyrosine also prevents the depletion of central and peripheral catecholamines (dopamine, norepinephrine, epinephrine) induced by acute stress, thereby eliciting protective effects on behavioral and cardiovascular parameters in the body.

TYRAMINE

Tyramine is high-normal (>80th percentile), which is usually due to eating foods high in this trace amine (protein). Specifically, tyramine is found in aged, fermented cured or spoiled food where microbes with decarboxylase enzymes convert tyrosine to tyramine. These foods include aged cheeses, smoked fish, cured meats, wine, and some types of beer. In sensitive individuals, high tyramine ingestion can trigger migraines by causing blood vessel restriction and then rebound vasodilation (Burns and Kidron 2020). Additionally, tyramine can trigger norepinephrine release, thereby stimulating sympathetic nervous system and consequently increase blood pressure. Because of this sympathetic mechanism, symptoms of agitation, anxiety, rapid heartbeat, and headaches may be noted.

THERAPEUTIC CONSIDERATIONS: avoid tyramine high foods (<https://headaches.org/wp-content/uploads/2018/02/TyramineDiet.pdf>) and calm the sympathetic nervous system. Supplements such as SAME, magnesium, vitamin B2 may aid with promoting norepinephrine metabolism. Additionally, nervines, adaptogens, L-theanine, biofeedback and meditation may help quiet down the overactive sympathetic response.

DOPAMINE

Dopamine is within reference range. Dopamine improves attention, focus, and motivation, helps with decision making, modulates movement control, promotes lactation, increases blood pressure, urine output and sodium excretion, and allows for feelings of reward and pleasure. Additionally, dopamine plays a central role in the etiology of addiction. Dopamine also serves as the parent precursor to norepinephrine and

epinephrine.

DOPAC AND HVA

DOPAC and HVA are above the optimal range, which is most likely due to use of a bioflavonoid supplement (e.g. quercetin, resveratrol, etc.), however this type of supplement was not listed on the requisition form. Research shows that intestinal bacteria break down quercetin and related flavonoids to liberate a variety of metabolites, including DOPAC and HVA (Weldin, et. al. 2003).

DOPAC and HVA are dopamine metabolites. Research shows that DOPAC and HVA are elevated in patients with anorexia nervosa (Van Binsbergen et al., 1991). HVA can also be elevated in lead toxicity (Tang et al., 1995), obsessive compulsive disorder (de Groot et al., 1995), and stress (Frankenhaeuser et al., 1986). In rare cases, HVA is elevated in patients with carcinoid tumors and pheochromocytomas (Davidson, 2005).

TREATMENT CONSIDERATIONS: Evaluate supplementation for bioflavonoid content. Therapies that help slow the MAO enzyme may also be helpful. These therapies include but are not limited to passiflora, curcumin, Huperzine A, and kava. Imaging may be appropriate in some patients.

NOREPINEPHRINE

Norepinephrine is within reference range. Norepinephrine functions both as a neurotransmitter and a hormone, participating in the body's "fight or flight" response. Norepinephrine increases alertness, focuses attention, fine-tunes vigilance, increases blood pressure, heart rate, and blood sugar, reduces digestive activity, pain, and sleep, prevents bladder emptying, and regulates body temperature. Norepinephrine is very similar in structure and physiological effects to epinephrine. The adrenal gland produces approximately 20% of the total output with 80% produced by the sympathetic nerve fibers.

NORMETANEPHRINE

Normetanephrine is low-normal (<20th percentile). Lower normetanephrine levels may be reflective of insufficient norepinephrine levels in the adrenal glands.

EPINEPHRINE

Epinephrine is within reference range. Epinephrine, also called adrenaline, functions both as a neurotransmitter and a hormone, participating in the body's fight or flight response. Epinephrine increases alertness, focuses attention, fine-tunes vigilance, increases blood pressure, heart rate, and blood glucose, reduces digestive activity, pain and sleep, prevents bladder emptying, and regulates body temperature.

VMA

Vanillylmandelic acid (VMA) is within reference range. VMA is a norepinephrine and epinephrine metabolite formed via the actions of monoamine oxidase, catechol-O-methyl transferase (COMT), and aldehyde dehydrogenase.

INFLAMMATORY MARKERS

KYNURENINE

Kynurenine is within reference range. Kynurenine is a central metabolite of the amino acid tryptophan with vasodilatory properties. Kynurenine is utilized by the body in the production of niacin (vitamin B3), eventually leading to the formation of NAD⁺, which plays a pivotal role in energy metabolism, gene expression, cell death and regulation of calcium homeostasis. More than 90% of the body's tryptophan is metabolized to the kynurenine pathway.

KYNURENIC ACID

Kynurenic acid is within range. Kynurenic acid is a neuroactive metabolite produced from kynurenine. Kynurenine is formed from tryptophan via the enzyme tryptophan dioxygenase and indoleamine 2,3-dioxygenase; and metabolized along two independent pathways to produce kynurenic acid via aminotransferases and 3-OH kynurenine.

Kynurenic acid (unless in excess amounts) is regarded to have a neuroprotective role because it inhibits the N-methyl-d-aspartate (NMDA) glutamate receptor, reduces the neurotransmitter glutamate release and thereby prevents excitotoxicity.

3-HYDROXYKYNURENINE

3-Hydroxykynurenine is within range. 3-Hydroxy Kynurenine (3-OH Kynurenine) is a metabolic intermediate of the kynurenine pathway, one of the major metabolites of tryptophan degradation. Kynurenine is transformed into 3-OH Kynurenine, which acts as a N-methyl-d-aspartate (NMDA) glutamate receptor agonist and has been demonstrated to exert neurotoxic effects, especially when in excess.

XANTHURENIC ACID

Xanthurenic acid is within range. Xanthurenic acid is a metabolite of the kynurenine pathway, formed directly from 3-OH Kynurenine, and serves as an indirect marker of vitamin B6 status.

Creatinine levels reflect urine concentration.

Low values suggest overly dilute urine; High values suggest overly concentrated urine.

Extreme low or high values may be induced by kidney or other metabolic disorders, but most values will be due to inadequate hydration (high creatinine) or excessive water intake in the several hours prior to testing (low creatinine). Creatinine is used to adjust the lab results for kidney function. No samples were refused due to quality issues.